

Hiroataka Hiraki

Research Scientist

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I am a Research Scientist on the HCI team at SB Intuitions, a SoftBank company focused on generative AI research and development. My research focuses on **speech interfaces and human-computer interaction**, particularly **speech input through very quiet utterances and in noisy environments**. I combine hardware prototyping, embedded systems, signal processing, and machine learning to develop speech interfaces for everyday use, including silent and whispered speech.

Education

- 2022.04–2026.09 **Ph.D. in Interdisciplinary Information Studies**, *The University of Tokyo*
Graduate School of Interdisciplinary Information Studies.
Thesis: Research on Wearable Speech Interfaces in Everyday Environments.
Advisors: Jun Rekimoto and Yoshio Ishiguro.
- 2025.12–2026.05 **Research Scholar (Internship)**, *University of Washington*
Paul G. Allen School of Computer Science & Engineering.
Advisor: Shyam Gollakota.
- 2020–2022 **Master's degree in Interdisciplinary Information Studies**, *The University of Tokyo*
Applied Computer Science.
Thesis: Research on Mask-type Wearable Silent Speech Interfaces.
Advisor: Jun Rekimoto.
- 2016–2020 **B.Eng., Information and Communication Engineering**, *The University of Tokyo*
Thesis: Recovery method of execution environment infected with ransomware by analyzing time series data of system calls.
Advisors: Hiroshi Esaki and Hideya Ochiai.
- 2012–2015 **Musashi High School**, *Tokyo, Japan*

Research and Professional Experience

- 2026.09–Present **Research Scientist**, *SB Intuitions*
HCI team.
- 2026.04–Present **Individual Researcher**, *JST ACT-X Acceleration Phase*
Wearable voice input/output interfaces for enhancing face-to-face vocalization.
- 2025.04–Present **Technical Advisor**, *Tokyo Ishigaku Sogo Kenkyujo, K.K.*
- 2023.10–2026.03 **Individual Researcher (Principal Investigator)**, *JST ACT-X*
Development of a wearable voice input/output interface for enhancing face-to-face vocalization.
- 2022.04–2025.03 **Research Assistant**, *AIST*
Human Augmentation Research Center. Sensing, signal processing, and wearable speech interfaces.
- 2022–2026 **Lead Engineer**, *AquaPlanet Inc.*
- 2021.07–2022.03 **Creator**, *IPA MITOU Program*
Wearable 3D pose-estimation devices using Unity and deep learning.
- 2017.08–2021.09 **Software Engineer**, *Yoshidumi Inc.*
Backend development using GCP, Java, and Go; web writing.

Peer-Reviewed Journal and Conference Papers

- 2025 **Hirotaka Hiraki**, Shusuke Kanazawa, Takahiro Miura, Manabu Yoshida, Masaaki Mochimaru, and Jun Rekimoto. "ClothTalk: Conductive Cloth Microphone for GPU-free, Real-time Clear Talk in Noisy Environments", in IEEE Access, vol. 13, pp. 199006-199023, 2025.
- 2025 **Hirotaka Hiraki** and Jun Rekimoto. "SilentWhisper: Enabling User-Independent Ultra-Low Volume Whispered Speech Recognition," IEEE Access, vol. 13, pp. 168782-168791, 2025.
- 2025 **Hirotaka Hiraki** and Jun Rekimoto. MaskClip: Detachable Clip-On Piezoelectric Sensing of Mask Surface Vibrations for Real-Time Noise-Robust Speech Input. In Proceedings of the Augmented Humans International Conference 2025 (AHs '25). 1-12.
- 2024 **Hirotaka Hiraki**, Shusuke Kanazawa, Takahiro Miura, Manabu Yoshida, Masaaki Mochimaru, and Jun Rekimoto. WhisperMask: a noise suppressive mask-type microphone for whisper speech. In Proceedings of the Augmented Humans International Conference 2024 (AHs '24). 1-14.
- 2024 Shota Imamura, **Hirotaka Hiraki**, and Jun Rekimoto. Serendipity Wall: A Discussion Support System Using Real-time Speech Recognition and Large Language Model. In Proceedings of the Augmented Humans International Conference 2024 (AHs '24). 237-247. **Best Paper Honorable Mention Award.**
- 2024 Takekazu Kitagishi, **Hirotaka Hiraki**, Hiromi Nakamura, Yoshio Ishiguro, and Jun Rekimoto. Pinching Tactile Display: A Cloth that Changes Tactile Sensation by Electrostatic Adsorption. In Proceedings of the 2024 International Conference on Advanced Visual Interfaces (AVI '24). Article 26, 1-9.
- 2022 Yusuke Kunimi, Masa Ogata, **Hirotaka Hiraki**, Motoshi Itagaki, Shusuke Kanazawa, and Masaaki Mochimaru. E-MASK: A Mask-Shaped Interface for Silent Speech Interaction with Flexible Strain Sensors. In Proceedings of the Augmented Humans International Conference 2022 (AHs '22). 26-34.
- 2021 **Hirotaka Hiraki** and Jun Rekimoto. SilentMask: Mask-type Silent Speech Interface with Measurement of Mouth Movement. In Augmented Humans Conference 2021 (AHs '21). 86-90.

Semi-Archival Peer-Reviewed Extended Abstracts

- 2025 **Hirotaka Hiraki** and Jun Rekimoto. SilentWhisper: inaudible faint whisper speech input for silent speech interaction. In Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems (CHI EA '25). Article 454, 1-5.
- 2024 **Hirotaka Hiraki** and Jun Rekimoto. Piezoelectric Sensing of Mask Surface Waves for Noise-Suppressive Speech Input. In The 37th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '24). Article 78, 1-3.
- 2024 **Hirotaka Hiraki**, Shusuke Kanazawa, Takahiro Miura, Manabu Yoshida, Masaaki Mochimaru, and Jun Rekimoto. Conductive Fabric Diaphragm for Noise-Suppressive Headset Microphone. In The 37th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '24). Article 56, 1-3.
- 2024 Naoto Nishida, **Hirotaka Hiraki**, Jun Rekimoto, and Yoshio Ishiguro. Real-Time Word-Level Temporal Segmentation in Streaming Speech Recognition. In The 37th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '24). Article 9, 1-3.

Extended Abstracts (continued)

- 2024 Shota Imamura, **Hirotaka Hiraki**, and Jun Rekimoto. Serendipity Wall: A discussion support system using real-time speech recognition and large language model. In 2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), pages 588-590, Los Alamitos, CA, USA, March 2024. IEEE Computer Society.
- 2024 T. Kitagishi, **H. Hiraki**, Y. Ishiguro, and J. Rekimoto. Pinching Tactile Display: A Remote Haptic Feedback System for Fabric Texture. In 2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), Orlando, FL, USA, 2024, pp. 594-596.
- 2023 **Hirotaka Hiraki**, Shusuke Kanazawa, Takahiro Miura, Manabu Yoshida, Masaaki Mochimaru, and Jun Rekimoto. External noise reduction using WhisperMask, a mask-type wearable microphone. In Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23). Article 454, 1-5.
- 2022 **Hirotaka Hiraki** and Jun Rekimoto. SilentWhisper: faint whisper speech using wearable microphone. In Adjunct Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct). Article 27, 1-3.

Workshop Paper

- 2021 Takuya Hara, **Hirotaka Hiraki**, Naoki Kimura, Hiromi Nakamura, and Jun Rekimoto. Pose-Tuner: Sonification Based Fine-Tuning of Pose with Unsupervised Learning. *Human Augmentation and Accessibility Workshop (HAA), CHI 2021*.
- Full publication list, including domestic papers: hirotakahiraki.github.io/publications/

Awards and Honors

- 2026.03 Information Processing Society of Japan (IPSJ) Yamashita Memorial Research Award
- 2025 CEATEC AWARD 2025 Digital Minister's Award
- 2025 IPSJ Interaction 2025 Best Paper, Best Presentation, and Interactive Presentation Awards
- 2025 Student Presentation Award (29th), Acoustical Society of Japan
- 2024 James Dyson Award 2024 Japan National Winner
- 2024 Best Paper Honorable Mention Award at Augmented Humans 2024 (AHs24)
- 2024 GUGEN 2024 Good Idea Award and two additional prizes
- 2024 IPSJ UBI Technical Committee #84 Student Outstanding Presentation Award
- 2022 Head of Department Award and Best Master's Thesis Award, Graduate School of Interdisciplinary Information Studies, The University of Tokyo
- 2022 Good Design New Hope Award, Honorable mention
- 2022 Next Age Award, Japan Interior Designers' Association (JID)
- 2022 Demo Award at Workshop on Interactive Systems and Software (Domestic Conference)
- 2021 Selected for the IPA MITOU Program
- 2021 Best Presentation Award at Silent Speech Recognition Workshop (Domestic Workshop)
- 2021 Best Presentation Award at Institute of Environmental Art and Design (Domestic Conference)
- 2021 Selected for the Hongo Tech Garage Spring Founder Program, The University of Tokyo

Grants and Scholarships

- 2026–2027 JST ACT-X Acceleration Phase
- 2025–2026 Tobitate! Study Abroad JAPAN, New Japan Representative Program (Innovator Course) USD 6,700.
- 2025–2026 Tateisi Science and Technology Foundation, International Exchange Grant (second round) USD 8,000.
- 2023–2026 JST ACT-X USD 34,000 over 2.5 years.
- 2022–2026 The University of Tokyo, International Graduate Program of Innovation for Intelligent World
- 2022 Japan Student Services Organization (JASSO) Scholarship. USD 6,500 awarded.
- 2021–2022 IPA MITOU Program USD 16,000 for one year.
- 2021 The University of Tokyo Ph.D. Fellowship. USD 9,000 over three years.

Teaching

- 2022–2023 **Teaching Assistant**, *The University of Tokyo*
Advanced Lectures in Human Augmentation Studies (GII-AC6312L2).

Technical Skills

- Programming Python, C/C++, C#, TypeScript, Ruby, R, Go, Java
- Web and cloud Ruby on Rails, Django, React, Apache, MySQL, GCP, AWS, Docker
- Machine learning PyTorch, TensorFlow, JAX
- Design Adobe Illustrator, InDesign, After Effects, Premiere Pro
- Prototyping Digital fabrication, Fusion 360, Rhinoceros, EAGLE, Unity

Academic Service

- 2025 Reviewer: TEI, Advanced Robotics, ICAT-EGVE, CISIM, and CHI Late-Breaking Work.
- 2024 Reviewer: CHI Late-Breaking Work.
- 2023 Reviewer: CHI Late-Breaking Work and MobileHCI Papers.
- 2022 Student Volunteer: CHI.